Brain & Language 276 (2026) 105737

Contents lists available at ScienceDirect

Brain and Language

journal homepage: www.elsevier.com/locate/b&l

Parallel contributions of socioeconomic status and bilingual experiences on executive function and the brain in young children

Gavkhar Abdurokhmonova [a,*], Ellie K. Taylor-Robinette [a], S. Alexa McDorman [a], Alexus G. Ramirez [a], Junaid S. Merchant [b], John D.E. Gabrieli [c], Rachel R. Romeo [a]

[a] Department of Human Development and Quantitative Methodology, University of Maryland, College Park, United States
[b] Department of Epidemiology and Biostatistics, University of Maryland, College Park, United States
[c] Department of Brain and Cognitive Sciences, Massachusetts Institute of Technology, United States

ARTICLE INFO

Keywords:
Bilingualism
Socioeconomic status
Executive functioning
fMRI

ABSTRACT

Executive functions (EF) are foundational for children's cognitive, socio-emotional, and academic outcomes, and socioeconomic status (SES) is consistently associated with disparities in children's EF performance and EF-related brain function. While bilingual children sometimes exhibit an "advantage" on certain EF tasks compared to their monolingual counterparts, it is unknown whether early bilingual experience may buffer the effects of low SES on EF in either brain or behavior. A SES-diverse sample ($n = 80$) of monolingual and bilingual children (4–6 years old) completed behavioral measures of EF and functional magnetic resonance imaging (fMRI) while engaging in a dichotic listening auditory selective attention task. Bilingualism was measured both by subjective parental report of children's language knowledge, as well as objective annotations of two daylong recordings of children's naturalistic language exposure. Selective auditory attention (vs. normal binaural speech) elicited activation in canonical attention/EF (posterior cingulate, precuneus, and cuneus) and language (posterior temporal gyrus and inferior frontal gyrus) regions. While there was no evidence of a protective effect of bilingualism, there were independent main effects of both SES and bilingual experience. Specifically, higher SES was associated with better EF performance and greater activation in bilateral precunei, independent of mono/bilingual status. While parent-reported bilingual knowledge was not associated with either brain or behavior independent of SES, children who had more balanced bilingual exposure at home exhibited better EF performance than those with more unbalanced or monolingual exposure in either language. Findings contribute to our understanding of the independent influences of specific experiences on children's EF development across contexts.

1. Introduction

Executive functions (EF) are a set of fundamental higher-level skills—including inhibition, cognitive flexibility, and working memory—that underlie goal-oriented behavior (Best & Miller, 2010; Miyake & Friedman, 2012). Largely supported by the late developing prefrontal cortex (PFC), EF skills provide the foundation for attention, planning, monitoring, and regulating behaviors and emotions (Moriguchi and Shinohara, 2019; Diamond, 2013; Durston et al., 2006)—all of which are critical for young children's academic success (Blair, 2016; Clark et al., 2010). Thus, it is not surprising that EF development is consistently predictive of academic performance, as well as many overall life outcomes (Ahmed et al., 2019; Burchinal et al., 2020).

The development of EF is a dynamic process, significantly shaped by both neurobiological maturation and environmental factors (Santa-Cruz & Rosas, 2017). EF skills develop significantly across the preschool years (3–6 years), largely supported by the maturation of frontal and parietal brain networks. The PFC, particularly the dorsolateral and anterior cingulate regions, shows significant structural and functional development during the preschool years (Brown & Jernigan, 2012), which is associated with marked improvements on tasks requiring working memory, inhibitory control, and cognitive flexibility (Buss & Spencer, 2014; Fiske et al., 2025). Similarly, the posterior parietal cortex (PPC) supports attention allocation, working memory, and decision-making, and greater recruitment of the PPC is linked with better performance on a variety of EF tasks in early childhood (Moriguchi and Shinohara,

* Corresponding author.
E-mail address: ga2541@umd.edu (G. Abdurokhmonova).

https://doi.org/10.1016/j.bandl.2026.105737
Received 14 August 2025; Received in revised form 15 February 2026; Accepted 1 March 2026
Available online 6 March 2026

2019). The protracted development of this fronto-parietal EF network throughout early childhood creates extended periods of experience-dependent plasticity, ultimately allowing for substantial individual differences in EF outcomes based on the variety of children's early experiences. This confluence of experience and the developmental timing of fronto-parietal brain maturation is presumably one reason that EF skills are one of the cognitive skills most affected by socioeconomic adversity (e.g., Merz et al., 2019).

Over recent decades, there have been a growing number of studies demonstrating significant associations between children's socioeconomic status (SES), EF development, and brain development (for review, see: Hackman & Farah, 2009 & 2010; Merz et al., 2019; Olson et al., 2021). Specifically, studies find that children growing up in lower SES environments are, on average, more likely to score lower on a variety of EF tasks (Lawson et al., 2018) and show differences in the structure and function of the PFC, as compared to their higher SES peers (Merz et al., 2019; Finn et al., 2017; Ursache and Noble, 2016; Ellwood-Lowe et al., 2022). Several mechanisms for this environmental effect have been suggested, including increased stress (Evans & Kim, 2013) and reduced cognitive stimulation (Rosen et al., 2019; Sheridan & McLaughlin, 2014), which alter the protracted development of the PFC in early childhood. Given the importance of EF skills for cognitive and socioemotional outcomes, it is critical to examine additional environmental factors that promote resilience to socioeconomic adversity and support EF development.

1.1. Bilingualism as a potential protective factor

The *Joint Attention Model* assumes that each language lexicon in bilinguals' minds is active during language use and competes for usage (Green, 1998; for a review, see Bialystok et al., 2009). Thus, constantly monitoring what language is relevant in a given context, inhibiting one language, and switching to another may allow for more efficiency and flexibility in switching between competing cognitive tasks (Bialystok, 2011; Poulin-Dubois et al., 2011), enhanced attention skills (Friensen et al., 2015), and more acute understanding of others' social intentions, perspectives, and desires (Goetz, 2003; Kovács, 2009). This is known as "bilingual advantage" (Bialystok et al., 2012). Green and Abutalebi (2013) underscored the importance of context in understanding the processes of language use in their *adaptive control hypothesis*, which proposes three possible contexts of language engagement: single-language, dual-language, and code-switching. Depending on which one of these three contexts bilinguals are engaged in, the languages may interact or compete. Therefore, each context is characterized by distinct behavioral and neurocognitive mechanisms underlying both language and EF networks.

Although there is a body of work showing that bilingual individuals perform better on EF tasks, recent meta-analyses find rather minimal or null evidence for its existence (Donnelly et al., 2019; Lowe et al., 2021; Paap et al., 2015; Paap & Greenberg, 2013). One potential reason for conflicting findings might be methodological and sampling differences across studies. The vast majority of studies of bilingualism and EF have been conducted in convenience samples of predominantly educated and economically advantaged participants from North America and Europe (Bialystok, 1999; 2011; Pelham & Abrams, 2014; Poulin-Dubois et al., 2011). Comparatively fewer studies have examined the existence of a bilingual advantage in lower SES children. In one study of low-SES second-graders in Portugal and Luxembourg, bilingual children outperformed monolingual children in measures of cognitive control (selective attention and interference suppression), but not abstract representation (abstract reasoning and working memory), suggesting that the benefits of bilingualism on certain EF measures extend to lower SES groups (Engel de Abreu et al., 2012). Similarly, in a study of low-SES Spanish-English third graders in the United States, the degree of bilingualism was associated with EF performance, such that children with more balanced language skills performed better on tasks of conflict resolution and working memory, suggesting additive benefits amidst the heterogeneity of bilingual development (Thomas-Sunesson et al., 2018).

Additionally, a few studies have compared EF skills between monolingual and bilingual children from various SES backgrounds. Carlson and Meltzoff (2008) compared low-SES Spanish-English Kindergarteners in the United States to higher-SES monolingual English Kindergarteners, finding that after controlling for SES, bilingual children performed better on EF tasks requiring the management of conflicting attentional demands, but not delay tasks. However, SES and EF were confounded in this case. In a factorial designed study, Calvo and Bialystok (2014) examined independent effects of SES and bilingualism on inhibition, working memory, and selective attention in a sample of 6-year-olds from working- and middle-class Canadian families. They found that, controlling for language status, middle-class children performed better than their working-class peers on all EF tasks, and controlling for SES, bilinguals performed better than monolinguals specifically on inhibition and working memory tasks. Another study of Italian preschoolers (with mixed second languages) also found significant independent effects of SES and bilingual exposure on working memory (Dicataldo & Roch, 2020).

While these studies provide some evidence on the independent effects of SES and bilingualism on EF development, they do not provide conclusive evidence on whether bilingualism and SES interact, such that bilingualism may *protect* against the negative effects of low SES on EF development. One study of British adults did find evidence of a protective effect, specifically for conflict resolution tasks (Naeem et al., 2018). However, whether this protective effect extends to children, as well as the neural mechanisms underlying EF development, are unknown.

1.2. Bilingualism and neurocognitive correlates of EF

A growing number of neuroimaging studies have examined differences in EF-related brain functioning between bilingual and monolingual children. In two studies using functional near-infrared spectroscopy (fNIRS), Spanish-English bilingual and English monolingual children performed similarly on an attentional control task, but bilinguals exhibited greater activation in left prefrontal and parietal regions (Arredondo et al., 2017; Arredondo et al., 2022). Further, the magnitude of left prefrontal activation during EF tasks scales with bilingual fluency (Arredondo et al., 2017; Xie et al., 2020), and even with the degree of bilingual language mixing heard by infants learning two languages (Arredondo et al., 2022). This suggests that both bilingual knowledge and day-to-day bilingual experiences can modulate neural processing of EF.

One fundamental component of EF that may be particularly supported by early bilingual environments is selective attention. Selective attention—the ability to focus on relevant signals and manage distraction—is linked to academic and socioemotional developmental outcomes (Garon et al., 2008; Stevens & Bavelier, 2012; Veer et al., 2017), and it undergoes rapid and extensive changes within the first few years of life (Blakey et al., 2016). In a recent electroencephalography (EEG) study of selective attention, Phelps et al. (2022) had children aged 7–12 years complete a dichotic listening task, in which they must attend to speech played into one ear while ignoring conflicting sounds played into the other ear. While both monolingual and bilingual children were equally good at inhibiting the distracting sounds in favor of the attended speech stream, bilinguals tended to show *weaker* neuronal encoding compared to monolinguals, especially during low interference conditions, thus exhibiting a redistribution of neurocognitive resources. This suggests a more *efficient* distribution of neuronal resources in bilingual children when encountering cognitively demanding tasks.

Together, these findings suggest that early bilingual experience may modulate the neural bases of selective attention. However, these studies do not address the broader socioeconomic context of participants. Further research is needed to understand how the proximal experience of bilingual exposure may interact with distal socioeconomic environments to shape brain functioning during selective attention processes.

1.3. The present study

The current study addresses these gaps by examining whether early bilingual experience may protect against the negative effects of low SES on EF and its neural bases in early childhood. Specifically, we examine independent effects of SES and bilingualism, as well as their interaction, on young children's behavioral EF skills and brain activation during an auditory selective attention task completed during functional magnetic resonance imaging (fMRI). This is the first study, to our knowledge, to explore relationships between SES, bilingualism, and the neurobiological bases of EF. Based on previous studies dichotic listening studies with adults (Hashimoto et al., 2000; Pugh et al., 1996), we expected to see increased activation during dichotic (versus binaural) listening in inferior frontal and superior temporal auditory/language areas, as well as prefrontal and superior parietal attention areas. Based on the behavioral findings linking SES, bilingualism, and EF (reviewed above), we hypothesized that higher SES and more frequent bilingual exposure would be associated with higher EF scores and greater brain activation in language and attention areas during the selective attention task. Finally, we hypothesize that bilingual status will moderate the relationship between SES and EF/neural activation during EF, such that greater bilingual exposure will protect against the negative effects of low SES on executive functioning and its neural underpinnings.

2. Methods and materials

A priori hypotheses and analyses were pre-registered at https://osf.io/sxkat. See below for variations from the pre-registration.

2.1. Participants

Participants were 80 children (47 male) between the ages of 4–7 years old (M = 5.82 years, SD = 0.64 years). Data collection took place in 2015–2016 in [BLINDED FOR PEER REVIEW]. Most participants (n = 52) were a subsample of a larger family-based intervention study (Romeo et al., 2021), which recruited primarily from lower SES oversubscribed charter schools and had no exclusion criteria other than being in the target age/grade range. The other 28 participants were recruited from the general public, with the only exclusion criterion being MRI incompatibility. All children with parent-reported SES and language status and valid EF measures from the baseline time point (before intervention randomization) were included.

See Table 1 for demographic measures. Parents reported whether children had any language, reading, learning, attention, mood, or cognitive difficulties. Most children (80%) were categorized as typically developing, while 20% (n = 16) were categorized as atypically developing for non-mutually-exclusive reasons including: a history of early intervention or having an individualized education plan (IEP; n = 10), born premature (n = 3), ADHD (n = 1), autism (n = 1), speech/language diagnosis (n = 9), repeating Kindergarten (n = 3), and/or clinically low standardized cognitive/language assessment scores (n = 8; note: all but one of these children had a reported diagnosis, and the other received a diagnosis after study conclusion). This distribution of typical and atypical development is representative of the general population, and thus all children were included in analyses with developmental status as a covariate. The monolingual and bilingual samples were matched on sex, age, developmental status, and EF scores, but the bilingual group had significantly lower income-to-needs ratio and marginally lower parent education as compared to the monolingual group; while the groups were well matched at the lower end of the SES distribution, there were more high-SES monolinguals than high-SES bilinguals. However, reported results do not change if conducted with a SES-matched subsample of participants (see supplement and discussion).

Table 1
Participant demographics and dependent measures for the full behavioral sample (n = 80).

Measure	Full Sample (n = 80)	Monolingual (n = 51)	Bilingual (n = 29)	Monolingual vs. Bilingual
Sex				p = 0.99
Male	n = 47 (59%)	n = 30 (59%)	n = 17 (59%)	
Female	n = 33 (41%)	n = 21 (41%)	n = 12 (41%)	
Age (years)	5.73 (0.69)	5.79 (0.68)	5.63 (0.73)	p = 0.33
Income-to-needs ratio	3.52 (2.51)	4.06 (2.57)	2.58 (2.13)	p = 0.008
Parent education (years)	14.83 (2.93)	15.4 (2.65)	14.00 (3.22)	p = 0.052
Race/Ethnicity				p = 0.006
White	n = 22 (28%)	n = 18 (35%)	n = 4 (14%)	
Black	n = 21 (26%)	n = 15 (29%)	n = 6 (21%)	
Hispanic	n = 22 (28%)	n = 8 (16%)	n = 14 (48%)	
Asian	n = 1 (1%)	n = 0 (0%)	n = 1 (3%)	
Other	n = 1 (1%)	n = 0 (0%)	n = 1 (3%)	
Multiple	n = 13 (16%)	n = 10 (20%)	n = 3 (1%)	
Development Status				p = 0.28
Typical	n = 64 (80%)	n = 42 (82%)	n = 22 (76%)	
Atypical	n = 16 (20%)	n = 9 (18%)	n = 9 (24%)	
EF Composite	0 (0.65)	0.04 (0.69)	−0.07 (0.59)	p = 0.90

Note: All continuous measures are presented as mean (standard deviation) and group comparisons are t-tests. Categorical measures are presented as count (percent) and group comparisons are chi-square tests.

Of the full sample, n = 44 (55%) had usable fMRI data (see below for exclusion criteria). The fMRI subsample (Table 2) differed from the full behavioral sample in that it had fewer atypically developing children (11%, including 1 with autism, 1 with ADHD, and 3 with a history of early intervention), fewer Hispanic participants (25%) and more White participants (43%), fewer bilingual participants (27%), and had a higher average SES. These demographic differences in behavioral vs. MRI data are also common in pediatric neuroimaging studies (Gard et al., 2023).

2.2. Family socioeconomic status

Participants were recruited to represent a wide socioeconomic range. Parents/guardians reported total family income in dollars, inclusive of wages, investments, child support/alimony, and state/federal aid, and the number of family members dependent on that income (inclusive of all adults and children). These were then converted into a federal income-to-needs (ITN) ratio, defined as the ratio of total income to the federal poverty level, such that ITN < 1 indicates below poverty and ITN > 1 indicates above poverty. ITN ranged from 0.24 to 10.29 (*M* = 3.52, *Med* = 2.84, *SD* = 2.51). Given the high cost of living in the region of data collection (relative to federal standards), participants were additionally classified by regional HUD Income Limits (https://www.huduser.gov/portal/datasets/il.html) into five ordinal bins: "extremely low income" (n = 15, 19%), "very low income" (n = 10, 13%), "low income" (n = 16, 20%), "below median income" (n = 13, 16%), and "above median income" (n = 26, 33%). Thus, two thirds of the sample lived below area median income (Fig. 1).

Parents also reported the highest level of education in years obtained by up to two parents/guardians, which were averaged together (*Range* = 7–21 years; *M* = 14.89, *Med* = 14.50, *SD* = 2.93). Parental education

Table 2
Participant demographics and dependent measures for the fMRI sample (n = 44).

Measure	Full Sample (n = 44)	Monolingual (n = 32)	Bilingual (n = 12)	Monolingual vs. Bilingual
Sex				p = 0.48
Male	n = 24 (55%)	n = 19 (59%)	n = 5 (42%)	
Female	n = 20 (45%)	n = 13 (41%)	n = 7 (58%)	
Age (years)	5.82 (0.64)	5.88 (0.63)	5.66 (0.65)	p = 0.34
Income-to-needs ratio	4.08 (2.82)	4.46 (2.84)	3.10 (2.63)	p = 0.15
Parent education (years)	2.61 (1.19)	2.73 (1.13)	2.29 (1.36)	p = 0.33
Race/Ethnicity				p = 0.47
White	n = 17 (39%)	n = 13 (40%)	n = 4 (34%)	
Black	n = 12 (27%)	n = 9 (28%)	n = 3 (25%)	
Hispanic	n = 8 (18%)	n = 5 (16%)	n = 3 (25%)	
Asian	n = 0 (0%)	n = 0 (0%)	n = 0 (0%)	
Other	n = 1 (2%)	n = 0 (0%)	n = 1 (8%)	
Multiple	n = 6 (14%)	n = 5 (16%)	n = 1 (8%)	
Development Status				p = 0.27
Typical	n = 39 (89%)	n = 28 (88%)	n = 11 (92%)	
Atypical	n = 6 (11%)	n = 4 (12%)	n = 1 (8%)	
EF Composite	0 (0.91)	0.27 (0.62)	0.21 (0.56)	p = 0.77

Note: All continuous measures are presented as mean (standard deviation) and group comparisons are t-tests. Categorical measures are presented as count (percent) and group comparisons are chi-square tests.

and ITN were highly correlated ($r = 0.71, p < 0.0001$). As pre-registered, we used ITN as our final measure of SES for all of our analyses. However, we also ran all ITN-involved analyses with SES composite of ITN and parent education (both averaged and z-scored). ITN and SES composite were highly correlated ($r = 0.93$; $p < 0.001$).

2.3. Bilingual characterization

A subjective rating of children's *bilingual knowledge* was determined by parent reported responses to two questions from the Language, Social, & Background Questionnaire (LSBQ; Luk & Bialystok, 2013): "Does your child understand/speak any language other than English? If so, please rate your child's understanding/speaking in the other language(s)", using a 5-point scale of "poor", "fair", "moderate", "good", "excellent". Children were categorized as monolingual (n = 48) if parents reported no understanding or speaking of other languages, or if both their understanding and speaking were rated Fair or less. Children were categorized as bilingual if either their understanding and/or speaking was rated "moderate" or better. The most frequent second language was Spanish (n = 21), though additional languages included Haitian (n = 9) and Cape Verdean (n = 2) Creoles, Mandarin (n = 3), Greek (n = 2), Russian (n = 1), German (n = 1), and Catalan (n = 1).

Additionally, children's objective *bilingual exposure* was measured by daylong audio recordings. Of the full n = 80 sample, n = 70 (88%) children completed 2 full days of Language and ENvironment Analysis (LENA) recording on non-school days. For all children whose parents reported at least some exposure to a second language, even minimally, their recordings were annotated for bilingual exposure using a python-based workflow (Cychosz et al., 2021). Trained research assistants annotated randomly sampled 30-second clips as to whether they contained only English, only non-English language(s), both English and non-English languages, or no speech. After reaching an asymptote (usually after an average of 120 clips), annotations were converted into the proportion of speech experienced in each category (e.g., 0.5 English, 0.3 Non-English, 0.2 Mixed). The outcome measure of interest was the absolute value of the difference between the English and non-English proportions (e.g., 0.2 in the example given), such that 0 indicated equal amounts of English and non-English exposure at home (i.e., balanced bilingual exposure), and 1 indicated either fully English or fully non-English exposure (monolingual). Participants whose parents reported no exposure to any languages other than English were automatically assigned a score of 1.

2.4. Executive function assessment

Children completed two behavioral assessments of EF skill. The Head-Toes-Knees Shoulders (HTKS, McClelland et al., 2014) task examines behavioral inhibition and cognitive flexibility for switching

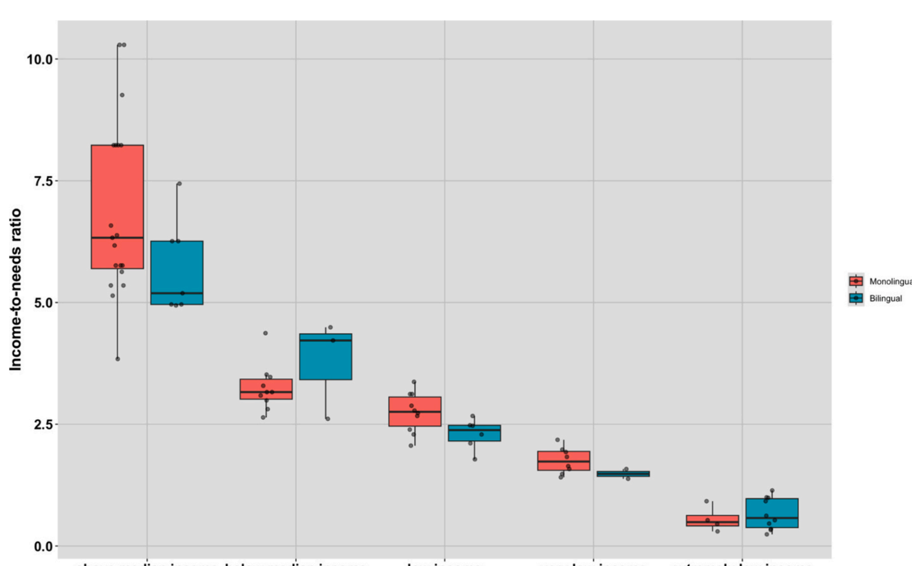

Fig. 1. Participants' family SES (measured by federal income-to-needs ratio) classified into 5 categories according to regional HUD Income Limits at the time of data collection.

rules. In the first part of the task, children were asked to do the opposite of a command (e.g., to touch their heads when the experimenter told them to touch their toes, and vice versa). In the second part, two additional commands were added touching shoulders and knees. In the final part, the rules were changed (i.e., now head goes with knees, and toes go with shoulders). The measure includes 30 items and children were given a score of 0 for an incorrect response, 1 for a self-correction (i.e., start the wrong movement, but change to the correct one), and 2 for a fully correct response, bringing the max score to 60.

The Hearts and Flowers (HF) task, a version of the dots task (Davidson et al., 2006) also examines inhibition, working memory, and cognitive flexibility using a touch screen computer. Stimulus images (either hearts or flowers) would appear on the left and right of the screen, and children were instructed to use their index fingers to touch buttons below the stimuli. After completing a touch-screen button practice, children completed three blocks: a congruent block of 12 trials, an incongruent block of 12 trials, and a mixed block of 49 trials. In all conditions, a red heart or flower appeared on the right or left side of the screen. In the congruent trials, children saw a heart and were asked to press the button on the same side as the heart. In the incongruent block, children saw a flower and were asked to press the button on the opposite side of the flower. In the mixed block, congruent and incongruent trials were intermixed such that children saw both hearts and flowers. The outcome measure of interest was the overall accuracy on the incongruent and mixed trials.

Overall accuracy on the HTKS and HF task were z-scored and averaged to form an executive functioning composite score, as in [BLINDED FOR PEER REVIEW].

2.5. fMRI neuroimaging procedure

All neuroimaging was conducted at [BLINDED FOR PEER REVIEW]. First, children were acclimated to the MRI environment and practiced lying still in a mock MRI scanner. Data were then acquired on a 3 Tesla Siemens MAGNETOM Trio Tim scanner (Siemens, Erlangen, Germany) equipped for echo-planar imaging with a 32-channel phased-array head coil. An automated scout image was acquired, and shimming procedures were performed to optimize field homogeneity. A whole-head, high-resolution T1-weighted multiecho magnetization-prepared rapid-acquisition gradient echo (MPRAGE) structural image was acquired using a protocol optimized for movement-prone pediatric populations [repetition time (TR) = 2,530 ms; echo times (TEs) = 1.64 ms, 3.5 ms, 5.36 ms, 7.22 ms; inversion time (TI) = 1,400 ms; flip angle = 7 degrees; resolution = 1-mm isotropic]. Whole-brain functional images were acquired with a continuous gradient echo-planar T2*-weighted sequence [T2*- weighted images, TR = 2,500 ms, TE = 30 ms, flip angle = 90 degrees, bandwidth = 2,298 Hz/Px, echo spacing = 0.5 ms, 41 transverse slices with field of view = 192 x 192, in-plane resolution = 3 mm x 3 mm]. Before each scan, six dummy volumes were acquired and discarded to reach equilibrium, and online prospective acquisition correction was applied to the echo-planar image sequence throughout the scan.

2.6. fMRI task

Children passively listened to short, simple stories derived from the Narrative Language Measures (NLM; Petersen & Spencer, 2012). All 40 stories from the NLM were edited to be exactly 4 sentences long with identical setup, problem, and resolution structure. The content of the stories included events that young children are likely to be familiar with, such as playing games/sports, eating meals, etc. Stories were then evaluated by two trained undergraduate research assistants who independently rated their level of engagement and suitability for preschoolers from diverse backgrounds on a scale from 0 to 5, and their scores were averaged. The 36 stories with the highest ratings (minimum 3.5 and higher) were selected and divided into two stimulus orders, including 6 presented binaurally, and 12 combined into dichotic pairs. The dichotic story pairs were matched such that the attended stories included equal numbers of male and female characters. All stories had consistent narrative structure, word count, and language complexity and were recorded by a female native-English speaker. Stimuli are available online at (https://osf.io/sxkat).

A block design paradigm presented 15-s-long trials consisting of an auditory story played binaurally (normally in both ears) or in a dichotic condition. A third condition (not analyzed here) involved playing the binaural stories backwards (see Romeo et al., 2018 for this condition). In the dichotic condition, a different story was played in each ear, one by the same female speaker as the other conditions, and one by a male speaker. The child was instructed to attend to the female voice and ignore the male voice, and they saw a visual reminder of a pink, female stick figure on the same side of the screen as the attended ear. The same stick figure appeared in the middle of the screen for binaural trials. Children practiced 1 binaural and 3 dichotic trials before entering the MRI, and were required to respond correctly to 2 of 3 dichotic condition comprehension questions to ensure understanding. One run consisted of 6 trials of each condition (18 trials total), each followed by a 5-second silent rest viewing a crosshair, such that the full run lasted 6 min total. Condition order was pseudorandomized so the same condition was never repeated twice consecutively. Each participant was randomly assigned to hear one of two stimulus lists containing all different stories with equal story interest ratings.

No real-time comprehension measure was administered, because this would have required much greater task demands on young participants (i.e., pushing a button, or speaking aloud) that would likely increase motion, and would have greatly increased scan time, also risking motion. During the piloting stage, researchers probed children to recall aspects of the cued stories after exiting the scanner. However, children could not recall much of either dichotic or binaural stories they heard in the scanner due to sufficient time having passed between the two events. Thus, no out-of-scanner comprehension measure was collected. However, children were still told before entering the scanner that they would receive prizes for accurate story comprehension, to incentivize engagement; then after the scan was completed, children were told there would be no comprehension test and all were given prizes.

2.7. fMRI preprocessing and analysis

Anatomical and functional images were preprocessed using fmriprep (version 20.2.5; Estaban et al., 2019). Structural data underwent intensity non-uniformity correction, brain extraction, and normalization to the ICBM 152 linear template, while functional data were additionally slice-time corrected, denoised of motion artifacts using independent component analysis (ICA-AROMA; Pruim, Mennes, Buitelaar, & Beckmann, 2015), and smoothed using a 6 mm FWHM Gaussian kernel. Volumes with greater than 1 mm framewise displacement (FD) were censored, and participants with fewer than 70% usable volumes were excluded (N = 14). Subject-level models were constructed with fitlins (version 0.11.0), controlling for the 6 motion parameters (x, y, z rotation and translation) and anatomical CompCor noise components.

fMRI whole group-level brain analyses were conducted by using AFNI's (version 20.2.14) 3dMEMA program which accounts for within-subject variance, and corrections for multiple comparisons were estimated using the clustsim program ($p < 0.001$ voxelwise and $p < 0.05$ clusterwise). Additionally, binary masks were created for all significant clusters at the whole-brain level, and the average participant-level activation within each cluster was extracted for further statistical analysis (see below).

2.8. Statistical analysis

With exception of the whole-brain fMRI analyses, all statistical analyses were completed in R (version 2024.12.0). First, we report zero-order correlations between all variables of interest and covariates. Then, we constructed a series of multivariate models to estimate (1) the main effects of SES and bilingualism (both binary classification and continuous exposure measures) and (2) potential interactions between SES and bilingualism on (1) out-of-scanner behavioral EF performance and (2) brain activation during selective attention (dichotic > binaural contrast). All models controlled for child age, sex, and typical development status (binary 0 or 1), and all models with fMRI measures additionally controlled for average FD. While covarying for typical development status was initially pre-registered only as an exploratory sensitivity analysis, it was shown to be related to some outcome variables, so was included as a covariate in all models.

To evaluate main effects on behavioral EF performance, we estimated two multiple regression models with the EF composite as the dependent variable, SES as the first independent variable, and either binary bilingual status or continuous bilingual exposure as the second independent variable (plus covariates). We then added an interaction term between SES and the bilingualism measure to evaluate whether bilingualism moderated the SES-EF relationship.

We used a similar approach for brain activation. First, we estimated a whole-brain, group-level model (averaged across all participants) to reveal regions of activation differences between dichotic and binaural conditions. Then, to evaluate main effects on selective attention activation, we estimated two whole-brain models with SES and either binary bilingual status or continuous bilingual exposure (plus covariates). To evaluate potential SES*bilingualism interactions, we first extracted participant-level dichotic > binaural activation magnitudes in all clusters significant at the average group level. These were then combined into an overall average weighted by cluster size. We then estimated two multiple regression models with activation magnitude as the dependent variable, SES and bilingualism (binary or continuous) as independent variables, and their interaction (plus covariates).

2.9. Variations from Preregistration

Slight modifications were made to the originally pre-registered analytical plan. First, we originally proposed to conduct all analyses only on the subsample with usable fMRI data; however, we decided it would be more representative to include all participants with usable behavioral data in the behavioral analyses. Second, we did not include reaction time (RT) on the HF task in the EF composite, because RT was negatively correlated with accuracy (indicating a speed-accuracy trade-off), so including both would have diluted estimates of children's EF performance. Finally, we had originally proposed to only control for (a) typical developmental status as a sensitivity check; however, given its relation to EF outcomes, we chose to include it as a covariate in all analyses.

3. Results

3.1. Behavioral EF results

To begin, we ran zero-order correlations between all continuous variables of interest and covariates (see Table 3). ITN was highly positively correlated with EF composite ($r = 0.33$, $p < 0.001$) indicating that children from higher SES backgrounds performed better in EF tasks. Furthermore, children's age was associated with EF composite ($r = 0.38$, $p < 0.001$): older children tended to have higher EF scores compared to their younger peers. Additionally, we did not find significant links between bilingual exposure and child age ($r = -0.16$, $p > 0.05$), ITN ($r = 0.21$, $p > 0.05$), nor EF composite ($r = -0.015$, $p > 0.05$). There was also no significant association between child age and ITN, meaning that children from our sample were not from a specific SES background.

3.1.1. Associations between EF, SES, and bilingualism

There was a statistically significant positive main effect of SES on EF,

Table 3
Zero order correlations between all continuous variables of interest and covariates.

	Child age	ITN	EF composite	Bilingual Exposure
Child age	1			
Income to needs ratio (ITN)	−0.14	1		
EF composite	0.38***	0.33**	1	
Bilingual Exposure	−0.16	0.21	−0.015	1

Note: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

after controlling for either children's parent-reported bilingual knowledge ($b = 0.085$, $p < 0.001$) or their bilingual exposure ($b = 0.087$, $p < 0.001$; Fig. 2a), indicating that children from higher SES backgrounds overall exhibited better EF performance. While there was no significant main effect of bilingual knowledge on EF after controlling for SES ($b = 0.143$, $p > 0.05$), there was a main effect of bilingual exposure on EF ($b = -0.05$, $p < 0.05$; Fig. 2b), indicating that children who were more balanced in their language exposure scored higher on EF tasks.

3.1.2. SES-bilingualism interactions on EF

Next, interaction terms were added to the two regression models. However, there was no significant interaction between SES and language knowledge ($b = -0.025$, $p > 0.05$), nor between SES and language exposure ($b = 0.104$, $p > 0.05$).

3.2. fMRI results

3.2.1. Whole-brain regions of task activation

As a group, participants exhibited greater dichotic vs. binaural activation in 5 clusters that span regions responsible for language processing (bilateral superior temporal and supramarginal gyri) and attention (right cuneus and precuneus, right inferior frontal gyrus, right middle cingulate cortex; see Fig. 3 for activation map and Table 4 for voxel wise activation in each cluster). Participant level activation in these clusters were extracted for further analysis (see below). Additionally, one cluster exhibited greater binaural vs. dichotic activation in the right inferior occipital gyrus (Table 4).

3.2.2. Whole-brain associations with SES and bilingualism

Next, we examined independent relationships between SES and bilingualism (both binary categories and continuous exposure measure) with activation in the dichotic > binaural contrast. SES (controlling for binary bilingual status) was significantly positively correlated with activation in a single cluster spanning left and right precunei (Fig. 4 and Table 5), indicating that higher SES was associated with greater selective attention-related activation in this posterior attention region independent of bilingual status. Results controlling for the continuous measure of bilingual exposure were very similar. However, neither the categorical nor continuous measures of bilingualism were associated with any significant clusters after controlling for SES.

3.2.3. SES-bilingualism interactions in selective attention clusters

Finally, we extracted participant-level average activations (i.e., regions of interest) across the 5 clusters found at the group level and created a composite weighted by cluster size. This composite was entered as the dependent variable in two separate multivariate regression models, each with SES as the first independent variable and bilingualism (binary classification or continuous exposure) as the second independent variable, plus the interaction between SES and bilingualism. Commensurate with the whole brain models, there was a main effect of SES ($b = 0.02$, $p < 0.01$) on the composite activation, such that children from higher SES backgrounds exhibited higher average activation across the 5 clusters. However, there was no main effect of language background (nor language knowledge: $b = 0.063$, $p > 0.05$; or

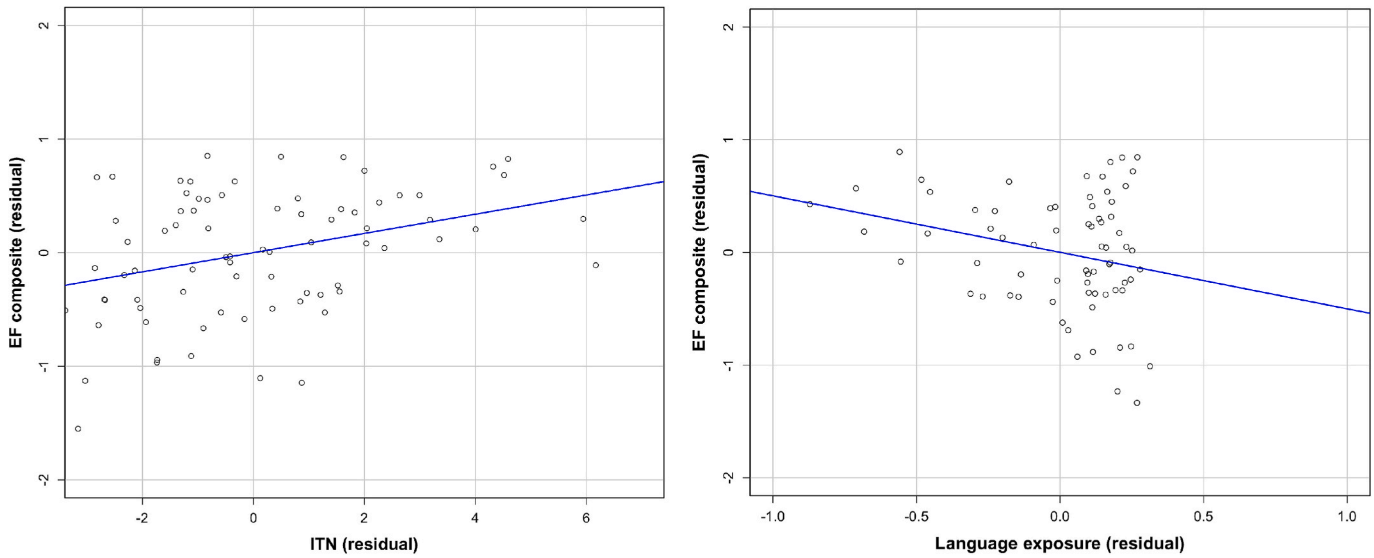

Fig. 2. Partial residual plots showing the main effects of SES (left; Fig. 2a) and language exposure (right; Fig. 2b) on EF, controlling for one another additionally to child's age, sex and developmental status. Higher SES (left) and more balanced bilingual exposure at home (right) are independently related to higher EF scores.

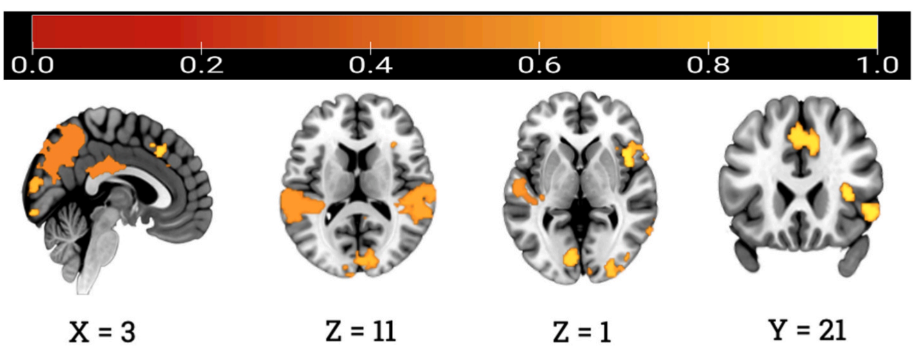

Fig. 3. Significant clusters emerging from a selective attention task (dichotic > binaural speech), averaged across all participants. Clusters span right cuneus and precuneus, bilateral superior temporal and supramarginal gyri, right middle cingulate cortex, and right inferior frontal gyrus.

Table 4
Significant clusters emerging from group level analyses of the selective attention task.

Cluster	# Voxels	CM x	CM y	CM z	Peak x	Peak y	Peak z
Dichotic > Binaural							
1. Right Cuneus and Precuneus	4191	−1.6	+68.1	+24.3	−4.0	+82.0	+44.0
2. Left Superior Temporal & Left Supramarginal Gyri	1441	+55.3	+33.1	+13.5	+68.0	+24.0	+14.0
3. Right Superior Temporal & Right Supramarginal Gyri	675	−56.6	+30.6	+13.4	−70.0	+22.0	+14.0
4. Right Middle Cingulate Cortex	266	−2.1	−20.6	+43.1	−2.0	−20.0	+40.0
5. Right Inferior Frontal Gyrus	222	−48.5	−16.2	−2.1	−56.0	−20.0	−4.0
Binaural > Dichotic							
1. Right Inferior Occipital Gyrus	383	−40.3	+77.9	−5.2	−36.0	+96.0	−2.0

Note: CM = Center of mass.

exposure: b = -0.05, $p > 0.05$) on composite activation. The interaction term was not significant in either model (SES*language knowledge: b = 0.01, $p > 0.05$ [Fig. 5]; SES*language exposure: b = -0.012, $p > 0.05$). Exploratory models treating each cluster separately exhibited the same lack of an interaction.

As exploratory analyses, we also created a SES composite variable by averaging and z-scoring ITN ratio and average parental education. We then ran all SES-involved analyses with this SES composite. The results were very similar to using ITN alone, with minor differences: controlling for bilingual status, SES was significantly associated with bilateral precunei (as found above), as well as two additional clusters: left and right parietal/superior lobules, which were sub-threshold in the ITN-only analysis (Supplemental Table 4 and Supplemental Fig. S1).

4. Discussion

The present study investigated the independent and interacting effects of SES and bilingual experiences (both bilingual knowledge and bilingual exposure) on the behavioral and brain bases of EF in 4–6-year-old children. Independent main effects of SES and bilingual experiences were found both in brain and behavior analyses. Specifically, independent of bilingual status, higher SES (measured as income-to-needs ratio) was associated with better EF task performance, as well as greater activation in bilateral precunei (medial superior parietal lobe) during an

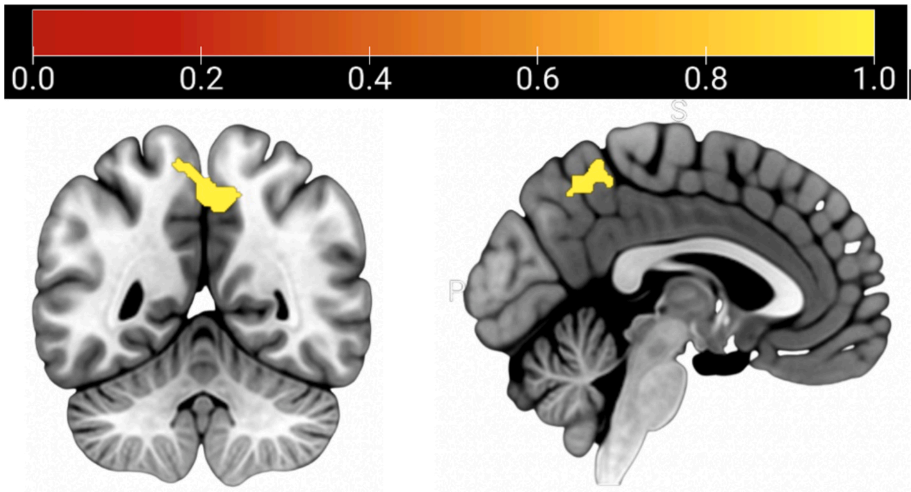

Fig. 4. Selective attention-related activation associated with SES after controlling for categorical bilingual status. A single cluster spanned bilateral precunei.

Table 5
Cluster where SES is significantly associated with selective attention activation (dichotic > binaural speech), controlling for categorical bilingual status.

Cluster	# Voxels	CM x	CM y	CM z	Peak x	Peak y	Peak z
Bilateral Precunei	308	−2.1	+52.3	+51.3	−2.0	+54.0	+52.0

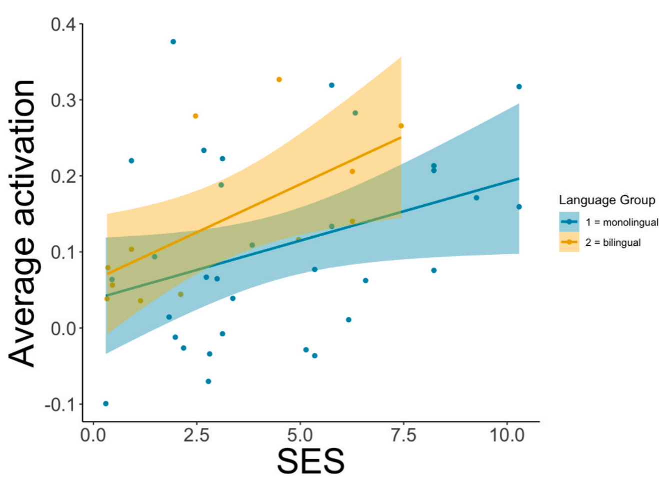

Fig. 5. The relationship between SES and selective attention brain activation is not moderated by bilingualism.

auditory selective attention task. Additionally, after controlling for SES, subjectively reported bilingual *knowledge* was not associated with either brain or behavior, but the objectively measured degree of bilingual *exposure* was correlated with EF task performance, such that more balanced bilingual exposure (i.e., equal exposure to two and/or more languages at home) was associated with better EF skills. Contrary to predictions, there were no significant interactions between SES and bilingual experiences, indicating no protective effect of bilingualism in the present case. However, findings contribute to our understanding of the independent influences of specific environments and experiences on children's EF development across contexts.

Our finding that SES is positively related to children's EF performance in brain and behavior is consistent with a vast prior literature. Children from lower SES backgrounds tend to score lower on assessments of EF-related functions, including inhibitory control, attention, working memory, and flexibility, all of which were required for the complex behavioral tasks in the present study (see Hackman et al., 2015 and Lawson et al., 2017 for review). Similarly, we found that lower SES was associated with reduced activation during the auditory selective attention task (relative to normal binaural listening) in bilateral precunei – a region in the medial superior parietal lobe and part of the dorsal attention network critically responsible for shifting attention between objects, features, or states, especially when the shift involves visuospatial processes (Yantis, 2008). While this was a purely auditory task, attention was shifted between the left and right ear, similar to canonical spatial auditory attention paradigms performed in freefield space (Coch et al., 2005). Combined with the additional visual cue as to which ear to attend, this could have recruited spatial attention processes to complete the task.

The SES differences in activation are consistent with several event-related potential (ERP) studies of selective auditory attention in children and adolescents from diverse SES backgrounds. These studies find that lower SES children exhibit decreased neuronal *recruitment* while processing relevant input, and decreased neuronal *suppression* of irrelevant input (D'Angiulli et al., 2012; Stevens et al., 2009; Hampton Wray et al., 2017), as well as faster neural recovery after suppressing irrelevant input (Stevens et al., 2015), even when attention allocation accuracy was equivalent to higher SES children (D'Angiulli et al., 2008). Further, amongst lower SES children, the degree of neurophysiological suppression of distractors is correlated with stress physiology (D'Angiulli et al., 2012) and nonverbal IQ (Isbell et al., 2016). Recently, this has been interpreted as "adaptive over-vigilance" by lower SES children, such that irrelevant information may soon become relevant and thus should not be ignored (D'Angiulli et al., 2024). The present findings extend these neurophysiological findings to hemodynamic effects, and localize them to the same regions responsible for allocating attention that are seen in dichotic listening fMRI tasks with adults (Pugh et al., 1996; Hashimoto et al., 2000). Taken together, these findings indicate that while SES continues to be a significant player in developmental trajectories of early EF, SES-diverse children tend to recruit distinct neuronal resources when engaging in EF tasks, likely indicating adaptive, experience-driven neurocognitive mechanisms of early cognitive processing.

Contrary to hypotheses, the present study finds limited evidence of bilingualism protecting against the effects of low SES on the behavioral and neural bases of EF. However, we do find that while monolingual and bilingual children did not differ on EF performance after controlling for SES, the degree of bilingual exposure did relate to EF performance. Specifically, children with more balanced bilingual exposure (i.e., equally exposed to two or more languages at home) in our sample performed better in behavioral EF tasks than their peers who heard more of one language (English or non-English) in their home environments. This suggests that regularly *experiencing* more than one language might be key to supporting better EF skills. Specifically, the need to frequently switch between processing different languages could hone children's ability to flexibly manage competing streams of information and inhibit distractors to focusing on target input. Regularly navigating a bilingual environment may provide more daily practice in these switching and inhibition skills. It is important to note that the degree of children's bilingual exposure was determined solely based on LENA data obtained from 2 full consecutive weekend days of home audio recordings when children were not at school. It is, however, possible that children heard more English at school or other community settings outside home.

Additionally, from a methodological perspective, these findings reveal the importance of considering more objective and naturalistic approaches to measuring children's language experiences. While caregiver surveys are a quick and feasible measure of children's early environments, "big data" approaches such as daylong recordings may allow for more nuanced investigations of important features in children's daily environments. Interestingly, the positive effect of bilingual exposure was limited to the behavioral tasks, and was not seen in brain activation during auditory selective attention. One potential reason for this is that the auditory fMRI stimuli were all in English, and thus may not have triggered advantaged selective attention skills. It is possible that different patterns of EF-based neuronal recruitment might be seen if the task called upon the full range of bilinguals' language skills. Additionally, more bilingual (versus monolingual) participants had to be excluded from the fMRI task due to motion, which may have limited our ability to find an independent effect of bilingual exposure on brain activation for selective attention.

Importantly, we take care to not over-interpret the null findings of independent effects of bilingual knowledge or exposure on the neural bases of selective attention. Specifically, the parent report measure of bilingual knowledge is a subjective measure reflecting how parents perceive their bilingual child's understanding and speaking of language(s) other than English, as opposed to objectively capturing children's bilingual skills. Comparatively, while our measure of bilingual exposure served as an objective measure of children's bilingual experiences, the small sample size ($n = 12$) of bilingual children, who had both LENA and neuroimaging data, might have precluded us from finding statistically significant effects of bilingual exposure on brain bases of selective attention. Furthermore, these are only two features of a complex and multidimensional phenomenon (Bialystok, 2016; Luk & Bialystok, 2013; Wagner et al., 2022). Future research should examine how additional graded features of bilingualism—such as age of acquisition, frequency of use, and objectively evaluated proficiency—independently and jointly relate to the neural bases of selective attention and EF more broadly.

This study had some additional limitations. First, an important limitation is that there was no measure of children's focus on the target story during the dichotic condition or story comprehension in either condition, because the young age of the participants limited real-time button presses and post-scanner recall ability (see methods). Thus, we cannot explicitly rule out the interpretation that variations in activation are due to differences in general task engagement, attention, or comprehension, rather than specific executive processes. However, we think that an interpretation of findings based solely on lack of engagement is unlikely given children's successful practice of the task and the incentivization of their attention. Further, the increase in activation in language and attention brain regions during the dichotic (versus binaural) condition is commensurate with selective attention studies of adults (with fMRI) and children (with EEG) that did assess comprehension, so this is unlikely to have changed results. However, because this is an important limitation, fMRI findings should be interpreted with caution.

Furthermore, we relied on parent report for classification of typical/atypical development, because not all children may have had access to diagnostic services, especially those from lower SES backgrounds. However, undiagnosed developmental conditions may have influenced children's EF and brain outcomes. Future research may include formal diagnostic assessments as a part of participant characterization to ensure equitable access and ultimately clarify potential relationships more precisely. Finally, the range of non-English languages represented were relatively limited, so we cannot generalize findings to all multilingual populations. Future research should strive to include a greater variety of language combinations.

Additionally, in the present sample, bilingualism and SES were somewhat confounded, such that, on average, the bilingual children had lower income-to-needs and marginally lower parental education (i.e., there were few higher-SES bilingual participants). This is a common issue in U.S. based studies, because—compared to other industrialized countries—bilingualism and SES are closely intertwined due to immigration patterns (Hoff, 2018; National Academies of Sciences, and Engineering and Medicine, 2015). Thus, in the US, bilingual children are, on average, more likely to come from lower-SES backgrounds (Interagency Forum on Child and Family Statistics, 2013). This poses difficulties in examining independent effects of bilingualism and SES on behavioral and neurocognitive developmental trajectories in early childhood. While some studies vary SES and bilingualism factorially, to examine their unique contributions and potential interactions (Calvo & Bialystok, 2013), others statistically control for for SES, especially when population-representative samples are naturally unbalanced (Meir and Armon-Lotem, 2017). While our sample is unbalanced, this likely did not prevent the finding of a protective effect, since such an effect would only rely on the lower-SES participants. In fact, it actually makes the finding of the independent SES and bilingual exposure effects more compelling, because these variables are correlated, yet still show independent influences on cognition. Furthermore, a sensitivity analysis with SES-matched subsamples (removing $n = 7$ high SES monolinguals, see supplement) resulted in the same findings, further suggesting that this confound likely had little influence on the findings.

In conclusion, selective attention for speech in young children relies on similar language and attention regions in the brain as in adults, and this attention effect is attenuated in lower SES children. While bilingualism did not protect against the effects of low SES on neurobiological or behavioral EF measures, SES and naturalistic bilingual exposure were independently related to EF task performance, indicating distinct influences of early experiences. Notably, this suggests that regularly *experiencing* more than one language might train EF skills better than simply *knowing* two languages. This ultimately contributes to our understanding of the independent and interacting effects of early experiences on children's EF and the underlying neurobiological processes.

CRediT authorship contribution statement

Gavkhar Abdurokhmonova: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Ellie K. Taylor-Robinette:** Writing – review & editing, Methodology, Conceptualization. **S. Alexa McDorman:** Writing – review & editing, Resources. **Alexus G. Ramirez:** Writing – review & editing, Software, Resources. **Junaid S. Merchant:** Writing – original draft, Visualization, Validation, Project administration, Methodology, Conceptualization. **John D.E. Gabrieli:** Supervision, Resources, Project administration, Data curation. **Rachel R. Romeo:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank Language, Experience, and Development (LEAD) lab research assistants, Fatimaria Rosales-Lima, Nicole Ostria, and Maria M. Egerton for dedicating extensive hours to annotate LENA home audio recordings so that we could derive the continuous measure of children's language exposure. We also thank the participants and their families who contributed to this study.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bandl.2026.105737.

Data availability

I have shared the link to my data/code at the Attach Files step

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